

Date: 11/9/26.

CN Assignment of
Name: Yash-Raj
Roll-no: 24k-0737.

Section (B) → Numericals.

Q#1

Data:- Root DNS server: 60ms

TLD DNS server: 50ms

Authoritative DNS server: 40ms

No cached info

Resolution uses iterative queries.

Q#2

(a) Total-Resolution time:-

$T_{DNS} = T_{Root} + T_{TLD} + T_{Autho.}$

$T_{DNS} = 60 + 50 + 40$

$T_{DNS} = 150ms$

Ans.

(b) Time-Saved:-

Root & TLD cached

So, $T_{new} = 40ms$

Total - T_{new}

$150 - 40$

$T_{saved} = 110ms$

Ans.

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Q#2

Data:

TTL = 300 sec = 5 minutes

uncached resolution time = 150ms

cached lookup time = 2ms

first resolution & cached at 10:00:00 AM

Sol:-

(a) query at 10:03:00 = 3 mins = 180 seconds

entry valid or not

if valid how much time saved

As TTL = 300 seconds

180 < 300 (cache entry is valid)

without-cache:-

150ms

with cache:- 2ms

Traved = 150 - 2

[Traved = 148ms] Ans.

(b) 7 minutes = $7 \times 60 = 420$ seconds

TTL = 300

420 > 300

So, this entry is not valid

Ans.

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Q#3.

Data:

file-size = 12MB $\times 8 = 96\text{mb}$

link 1 = 20Mbps

link 2 = 100Mbps

1MB = 8 Megabit

Q1.

$L = 96\text{mb}$

link 1 =

$R_1 = 20\text{Mbps}$

$T_1 = \frac{96\text{mb}}{20\text{Mbps}}$

$\Rightarrow [T_1 = 4.8\text{s}]$

20Mbps

link 2 =

$R_2 = 100\text{Mbps}$

$T_2 = \frac{96\text{mb}}{100\text{Mbps}}$

$\Rightarrow [T_2 = 0.96\text{s}]$

100Mbps

Total transmission delay = $T_1 + T_2$

$4.8 + 0.96$

$[5.76\text{sec}]$

Ans.

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Q#41

158000 bits

link 1: 40Mbps, dprop = 20ms

link 2: 80Mbps, dprop = 30ms

link 3: 100Mbps, dprop = 5ms

End-to-End delay = ?

Soln

$$\text{link 1: } \frac{L}{R} = \frac{158000}{40 \times 10^6} = 3.95 \times 10^{-3} \text{ s} = 3.95 \text{ ms} \approx 4 \text{ ms}$$

Link 1 = 4 + 20 = 24 ms

$$\text{link 2: } \frac{L}{R} = \frac{158000}{80 \times 10^6} = 1.975 \times 10^{-3} \text{ s} = 1.975 \text{ ms} \approx 2 \text{ ms}$$

Link 2 = 2 + 30 = 32 ms

$$\text{link 3: } \frac{L}{R} = \frac{158000}{100 \times 10^6} = 1.58 \times 10^{-3} \text{ s} = 1.58 \text{ ms}$$

Link 3 = 1.58 + 5 = 6.58 ms

$$\text{End-to-End delay} = 24 + 32 + 6.58 = 62.58 \text{ ms}$$

Date: 11/9/21

Roll no: 24K-0737

Q. 15.

Data:-

Requests / hour = 1000

Avg - object size = 400kb

Cache hit-ratio = 75%

miss ratio = $1 - 0.75 = 0.25$

Soln

(a) Total traffic without caching:-

Traffic = $1000 \times 400\text{kb}$
= 400000kb.

1kb = 1000000kb.

~~25~~ 2Gb.

[Traffic = 2Gb/hour] Ans.

(b) Traffic reaching origin server after caching

$1000 \times 0.25 = 1250$

= $1250 \times 400\text{kb}$.

= 500,000kb.

= 0.5Gb

so, [0.5Gb/hour] Ans.

(c) Data traffic saved per hour :- 2Gb/hour (original).
after - caching (0.5Gb/hour)

2 - 0.5

[1.5Gb/hour]

Ans =